

Great Representation. Where Are the Brakes?

Why AI safety needs causal enforcement, not just knowledge of constraints.

Imagine entering a lift on the 60th floor.

PLAQUE, MOUNTED BESIDE THE DOOR

"This lift contains a state-of-the-art representation of braking behaviour."

I'm taking the stairs.

**I don't want the lift
to understand braking.
I want brakes.**

The lift may:

- Understand when braking is appropriate
- Model safe descent
- Recognise overspeed
- Predict failure

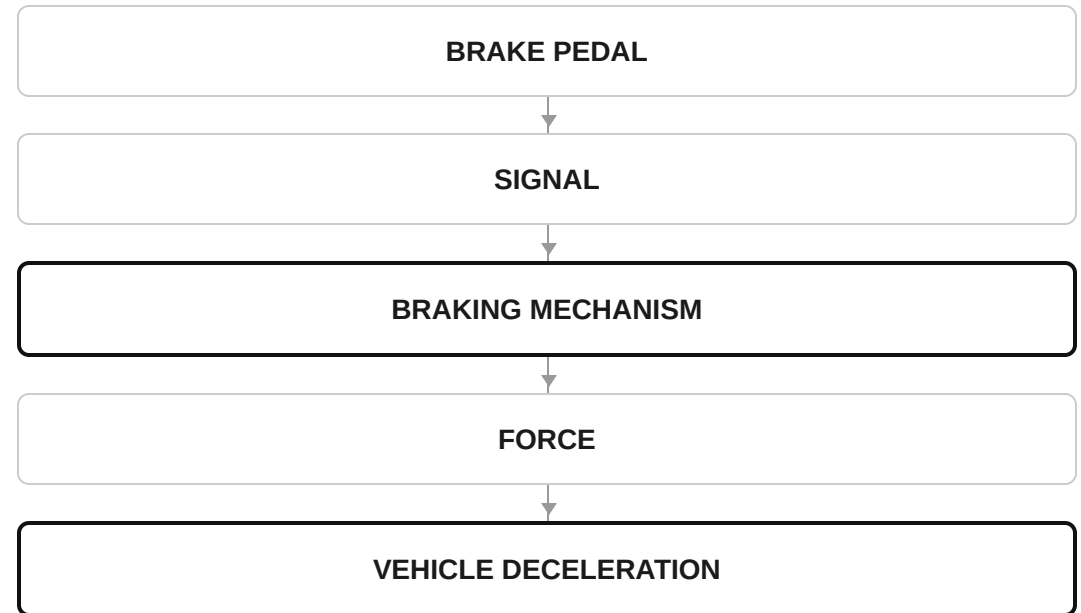
But none of those things physically stop the lift.

What mechanism changes the trajectory?

SALESMAN

"The AI has an exceptionally detailed representation of deceleration."

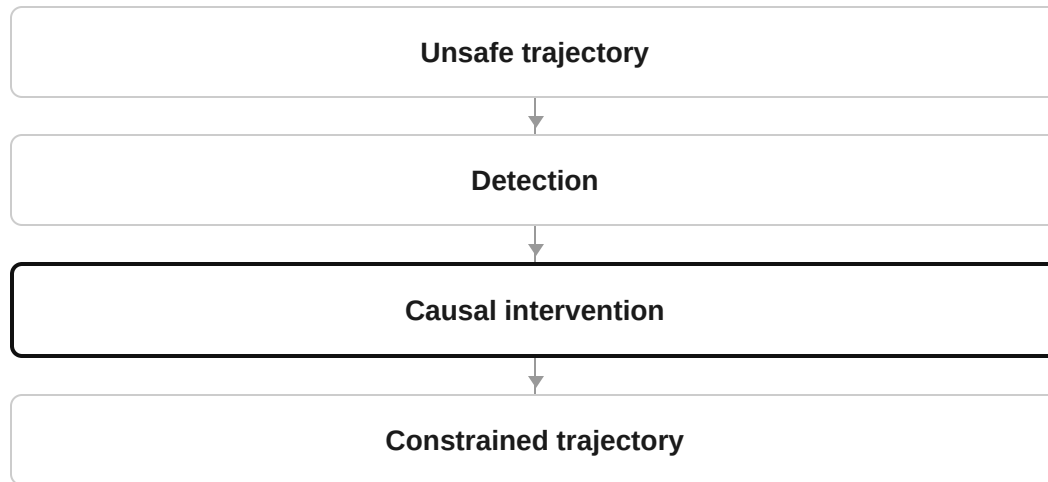
"When I press the pedal, do the calipers close?"



"Our newest vehicle achieved 99.8% seatbelt understanding."

Fantastic.

**Am I physically attached to the car —
or am I meeting the windscreen?**



**The system does not merely
know the boundary.**

Something enforces it.

- Call APIs
- Send communications
- Write and execute code
- Modify databases
- Trigger workflows
- Operate computers
- Move money
- Interact with physical systems

**And somehow one of our dominant safety ideas became:
Tell the model what it isn't allowed to do.**

Where are the brakes?

THE £80,000 EXAMPLE

POLICY

"Never transfer more than £10,000 without human approval."

The model can understand it, repeat it, explain it, and pass evaluations about it.

PROPOSED ACTION

TRANSFER £80,000

~~"Did the model understand the policy?"~~

WRONG QUESTION

"What stops the action from reaching the bank?"

THREE DIFFERENT CLAIMS

REPRESENTATION

The system knows the constraint.

BEHAVIOURAL COMPLIANCE

The system usually follows the constraint.

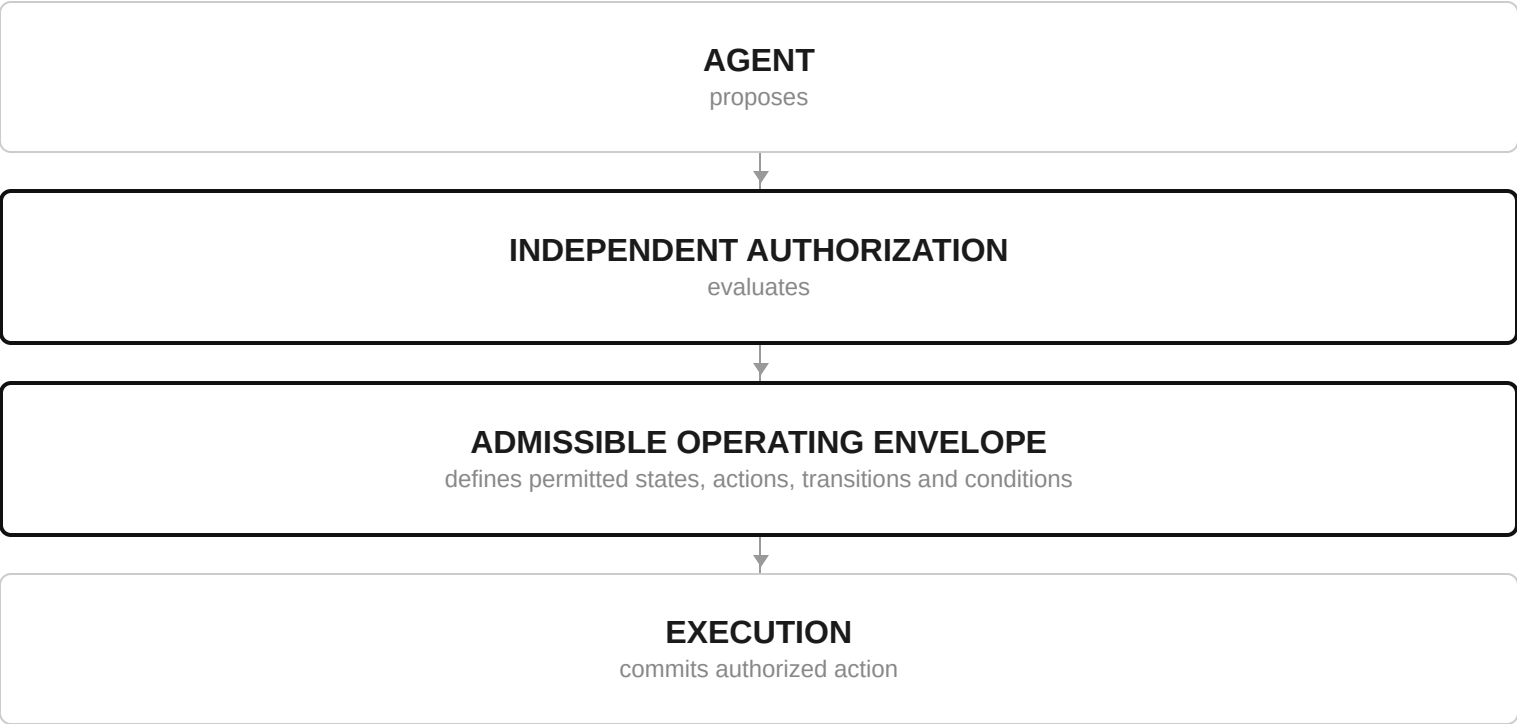
CAUSAL ENFORCEMENT

The architecture prevents prohibited execution.

These are not the same safety claim.

**Knowing where the boundary is
does not mean
you cannot cross it.**

Humans demonstrate this rather convincingly.



**The agent gets to keep
its intelligence.**

It does not automatically inherit unlimited authority.

"An Admissible Operating Envelope is the set of states, actions, transitions and operating conditions that a system is permitted to occupy or execute within a defined environment."

Given the current state and proposed transition:

Is the next state admissible?

"How often does the system avoid dangerous behaviour?"



**"Under this bounded environment and governance configuration,
can the prohibited state be reached at all?"**



unsafe branches terminate at the independent authorization boundary

This is bounded verification — not a claim of universal AI safety.

A hallucinated paragraph is one class of problem.

A hallucinated database deletion executed through an API is another.



*The more capable the agent becomes, the more
important it becomes to separate:*

CAPABILITY / AUTHORITY

*"Don't worry.
The lift has an excellent representation of braking."*

You're taking the stairs with me.

Intelligence can represent a boundary.

Alignment can encourage behaviour inside the boundary.

Evaluation can measure how often the boundary is respected.

But consequential autonomous systems also require mechanisms that causally govern which transitions are allowed to execute.

WHERE ARE THE BRAKES?

Representation $\neq$ causal enforcement

RESURRECTION TECH™